Rotating Equipment Reliability & Vibration Governance Manual

Document ID:	APEL-MAN-ENG-201	**Effective Date:**	2026-08-01
Revision:	4.0	**Classification:**	CONFIDENTIAL - STRICT PROPRIETARY
Department:	Mechanical Reliability Engineering	**Site Location:**	Jamnagar Refinery Complex (JRC)

1. Purpose and Scope

This authoritative engineering manual establishes rigorous operational, surveillance, and maintenance standards for rotating machinery across all ApexPetro production units (CDU-4, HCU-1, DISP-1, UTIL-1). Governs centrifugal pumps, compressors, and motor drives.

2. Vibration Acceptance Standards (ISO 10816-3 Class II)

- Zone A (0.1 - 2.8 mm/s RMS): Newly commissioned or completely overhauled machinery condition.
- Zone B (2.8 - 4.5 mm/s RMS): Continuous unrestricted long-term operation acceptable.
- Zone C (4.5 - 7.1 mm/s RMS): Advisory warning alert boundary; mandatory weekly spectrum analysis.
- Zone D (> 7.1 mm/s RMS): Unacceptable vibration; automatic or mandatory manual trip within 15 minutes.

3. Centrifugal Pump Operational Health Protocols (P-102 Case Baseline)

3.1 Mechanical Seal Plan 53B Pressurization

Pumps handling volatile crude or light hydrocarbons must maintain barrier fluid pressure at minimum 2.0 bar above maximum seal chamber pressure. Loss of barrier pressure triggers immediate alarm.

3.2 Casing Thickness Retiring Criteria

P-102 casing minimum allowable thickness under API 610 calculation is 3.5 mm. Thickness below 4.0 mm mandates maximum allowable working pressure (MAWP) reduction to 40.0 bar(g).

4. Compressor Anti-Surge & Dynamic Stability

Hydrogen recycle compressors (C-301) must maintain minimum 15% surge margin under all molecular weight excursions. Anti-surge fast-opening valves must stroke 0 to 100% within 1.2 seconds.